Bragg band gaps in a LEGO sonic crystal along ΓX and ΓM

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Abstract. A sonic crystal blocks sound in a band of frequencies, the band gap, expected near $f_B = c/(2d)$, with d the spacing between planes of scatterers. We measured the transmission of white noise (spectrum with crystal divided by spectrum without crystal) through square lattices of 2×2 LEGO columns, with lattice constant $a = 32$–56 mm and $N = 1$–11 rows, along ΓX and ΓM (lattice rotated by 45°), and compared it with a plane-wave expansion (PWE). Along ΓX the lower gap edge agrees with the PWE within 1.3σ for $a \leq 48$ mm but not for $a = 56$ mm (3.9σ); the gap does not scale as $1/a$ ($\chi^2/\nu = 15$), and the attenuation grows linearly with N, by 1.45 ± 0.29 to 1.98 ± 0.41 dB per row ($a = 40$–56 mm). Rotating the crystal to ΓM raises the lower edge by a factor 1.27–1.45, consistent with $\sqrt{2}$ within 2σ. The precision is limited by the reference measurement and by sound bypassing the crystal, and we find no evidence of a complete gap.

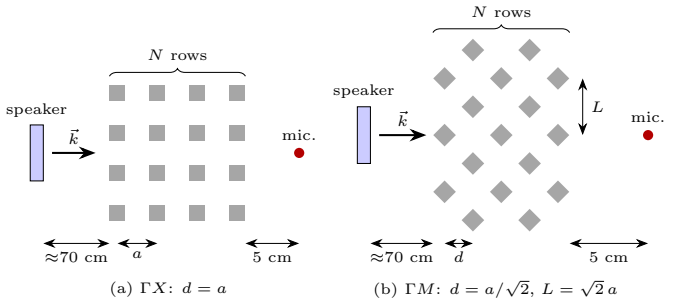


Figure 1. Set-up seen from above (not to scale). (a) ΓX: aligned columns. (b) ΓM: lattice rotated by 45°, with a brick vertex facing the loudspeaker and rows alternating between 4 and 3 bricks. a: distance between columns; d: distance between planes; L: period within a plane.

1 Introduction

A sonic crystal is a periodic array of rigid obstacles in air. Reflections from successive planes of obstacles, separated by d, add in phase when $\lambda = 2d$, and around

$$f_B = \frac{c}{2d}, \qquad d = a \ (\Gamma X), \qquad d = a/\sqrt{2} \ (\Gamma M) \tag{1}$$

a band gap opens: at the edge of the Brillouin zone, $k = \pi/d$, the standing waves with maximum pressure on the planes of bricks and in the air between them no longer have the same frequency, and between the two, $f_{\text{inf}} < f_B < f_{\text{sup}}$, no wave propagates. Inside the gap $k = \pi/d + i\kappa$ and the pressure decays as $e^{-\kappa x}$; after N rows the attenuation in decibels is

$$D(N) = \alpha N, \qquad \alpha = 20 \log_{10}(e)\, \kappa d. \tag{2}$$

The position and width of the gap depend on the filling fraction $ff = s^2/a^2$ (0.24 to 0.08 here): the bricks slow the sound down, $c_{\text{eff}} \simeq c/\sqrt{1+ff}$, and the gap moves below f_B, more so for smaller a. The bands are computed by solving $\nabla \cdot (\rho^{-1} \nabla p) + \omega^2 p/(\rho c^2) = 0$ with p written as a Bloch wave (PWE, Appendix B). We tested, along ΓX and ΓM, the gap edges, their dependence on a, the linear attenuation in N, and whether the gaps of the two directions overlap (complete gap).

2 Method

Experimental set-up. The crystal is a square lattice of columns (Fig. 1), each one a stack of 2×2 LEGO bricks (side $s = 15.8$ mm) on a LEGO baseplate. The lattice constant a is the centre-to-centre distance between neighbouring columns.

Along ΓX the rows are at $d = a$. Only this spacing along the propagation direction sets the Bragg frequency, $f_B = c/(2d)$, and it was the only one changed: the lateral spacing and the enclosure were kept fixed. Along ΓM the lattice is rotated by 45° and the two spacings can no longer be chosen separately: the planes are at $d = a/\sqrt{2}$ and the columns within a plane at $L = \sqrt{2}\, a = 2d$, so $f_B = c/L$. Increasing a therefore widens the lattice laterally as well.

The crystal is enclosed by LEGO walls and roof and placed in a cardboard box with foam around it. A Bluetooth loudspeaker ≈ 70 cm from the crystal face emits white noise, with frequencies between 500 and 12 000 Hz, for 10 s, and a USB microphone 5 cm behind the crystal records the sound. The loudspeaker distance satisfies the plane-wave criterion (Appendix B).

Parameters varied and repetitions. We varied $a = 32$, 40, 48 and 56 mm along ΓX and $a = 32$, 40 and 48 mm along ΓM, and the number of rows N, removing one row at a time from the microphone side ($N = 10 \to 1$ along ΓX and $N = 11 \to 1$ along ΓM). The loudspeaker, the microphone and the volume were kept fixed. Each configuration had 5 recordings, for a total of 72 configurations and 360 recordings, of which 16 were excluded for saturation or Bluetooth dropouts. The 4 sessions took place at $\theta = 20.5\,^\circ$C and $20.0\,^\circ$C ($\pm 1\,^\circ$C), which gives $c = (343.5 \pm 0.6)$ m/s and (343.2 ± 0.6) m/s.

Quantity measured and data processing. We measured the relative transmission

$$T(f) = 10 \log_{10}\left[\frac{\bar{P}_{\text{crystal}}(f)}{\bar{P}_{\text{ref}}(f)}\right] \ \text{(dB)}, \tag{3}$$

where P is the power spectral density of the recorded signal, and the reference is the same set-up without columns. The division removes the response of the loudspeaker, the microphone and the room.

Spectra. P was computed with Welch's method (Hann window, 4096 points, $\Delta f = 11.7$ Hz at a sampling rate of 48 kHz) on the segment 0.8–10.3 s of each recording, which excludes the Bluetooth delay.

Smoothing. To reduce the statistical fluctuations of the spectrum, which are amplified when dividing two spectra, each P was smoothed in power. The window is 1/6 of an octave wide (±6 % of f).

Reference. The reference was measured for each value of a at the start of the measurements. For each a we used, among the references of the session, the one that makes $T(f)$ flattest in the control band (see below); the others enter the uncertainty associated with the reference.

Control band. As a check of the measurement, T should be constant and close to 0 dB in the band 0.35–0.70 f_0, with $f_0 = c/(2a)$ along both directions. This band lies at frequencies below the first gap of the lattice, where the crystal is practically transparent.

3 Analysis

From each $T(f)$ we extract T_0 and the ripple (median and robust deviation of T in the control band, Sec. 2); the gap edges f_{inf} and f_{sup}, where T crosses $T_0 - 6$ dB around 0.75–1.25 f_B; and the depth $D = T_0 - \langle T \rangle_{f_B \pm 10\%}$. For each a, the edges are the weighted mean over $N \geq 4$; these are compared with the PWE and fitted against $1/(2d)$, and $D(N)$ is fitted with Eq. (2).

Uncertainties. Each number has four independent contributions, $\sigma^2 = \sigma_{\text{est}}^2 + \sigma_{\text{rip}}^2 + \sigma_{\text{ref}}^2 + \sigma_{\text{res}}^2$: σ_{est}, the standard error of the 5 repetitions (negligible); σ_{rip}, the ripple of T propagated to the number (0.92×ripple for D, and ripple/$|dT/df|$ for an edge); σ_{ref}, the spread of the result when it is computed with each reference of the same day; and, for frequencies only, the spectral resolution $\sigma_{\text{res}} = \Delta f/\sqrt{12} = 3.4$ Hz. Per block, $\sigma^2 = \sigma_{\text{disp}}^2 + \sigma_{\text{sys}}^2$. The systematic uncertainty (ripple and reference, common to the whole block) does not decrease with averaging and dominates: for f_{inf} at $a = 48$ mm, $\sigma_{\text{disp}} = 28$ Hz and $\sigma_{\text{sys}} = 98$ Hz. Fits are weighted least squares; if $\chi^2/\nu > 1$ the uncertainties are multiplied by $\sqrt{\chi^2/\nu}$. Comparisons use

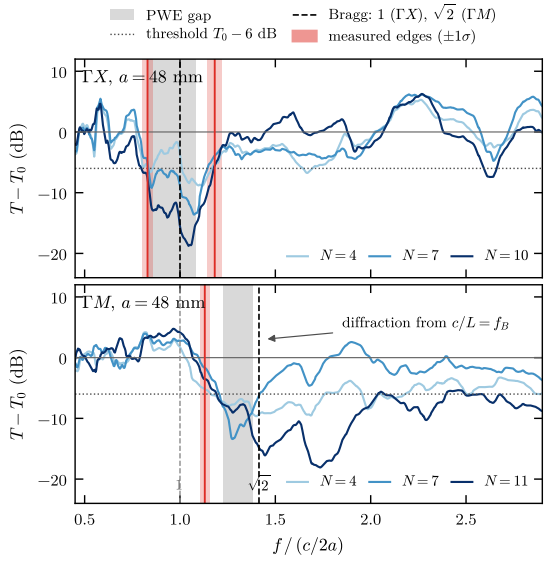


Figure 2. Relative transmission for $a = 48$ mm. Grey: PWE gap; red: measured edges (mean over $N \geq 4$). Along ΓX the gap closes; along ΓM lateral diffraction starts at f_B and T does not return to 0. All a in Appendix A.

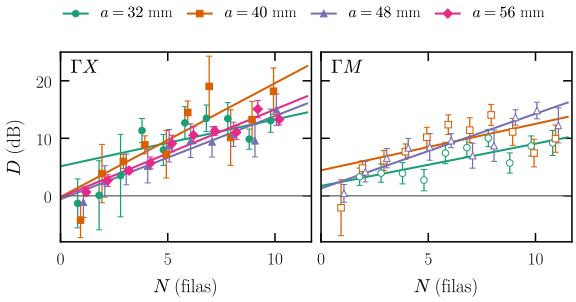


Figure 3. Depth D at $f_B \pm 10\,\%$ versus number of rows, with the line $D = \alpha N + b$ fitted for each a (slopes in Table 1).

$$z = |x - x_0| / \sqrt{\sigma_x^2 + \sigma_0^2}.$$

4 Results

Gap edges. Along ΓX there is a well-defined gap for $N \geq 4$ for all a (Fig. 2); it closes in 27 of the 28 measurements with $N \geq 4$. The edges are in Table 1: for $a \leq 48$ mm, $f_{\inf}$ agrees with the PWE within 1.3σ and $f_{\sup}$ within 2.6σ; for $a = 56$ mm both lie above ($z = 3.9$ and 3.6).

Dependence on a. The gap does not scale as $1/a$: f_c/f_B rises from 0.90(2) at $a = 32$ mm to 1.09(2) at 56 mm. A line $f_c = m/(2a)$ through the origin gives $m = 339 \pm 15$ m/s, compatible with c ($z = 0.3$) but with $\chi^2/\nu = 15$; the free line ($\chi^2/\nu = 1.4$) gives $m = 229 \pm 20$ m/s and 1280 ± 230 Hz.

Attenuation. Along ΓX the depth grows linearly with N (Fig. 3): for $a = 40$, 48 and 56 mm, the intercept $b = -0.2 \pm 2.0$, -0.6 ± 1.9 and -0.3 ± 0.6 dB which is compatible with 0, as Eq. (2) predicts. For $a = 32$ mm, D saturates at ≈ 13 dB from $N = 4$, which pulls the fitted line up ($b = 5.1 \pm 2.9$ dB). The measured α is 15, 50, 53 and 80 % of the PWE value for $a = 32$–56 mm.

Rotated crystal (ΓM). Rotating the lattice by 45° moves the gap up, as expected from the closer planes (Fig. 2). The lower edge rises by a factor $f_{\inf}(\Gamma M)/f_{\inf}(\Gamma X) = 1.45 \pm 0.08$, 1.27 ± 0.07 and 1.36 ± 0.06 for $a = 32$, 40 and 48 mm, consistent with the Bragg estimate $\sqrt{2}$ ($z \leq 2.0$) but below the PWE prediction of 1.59–1.47, because $f_{\inf}(\Gamma M)$ itself lies 8–11 % below the PWE ($z = 2.5$–3.6). Above f_B the gap rarely closes (12 of 24 measurements with $N \geq 4$). The overlap with the ΓX gap, $f_{\sup}(\Gamma X) - f_{\inf}(\Gamma M) = 420 \pm 330$, 560 ± 290 and 190 ± 160 Hz, is below 2σ, while the PWE predicts no overlap (-140 to -500 Hz).

Table 1. Results per block (mean over $N \geq 4$) versus the PWE. Frequencies in Hz; α in dB/row. † The gap does not close. ‡ Only 14–38 % of the $f_B \pm 10\,\%$ window lies inside the theoretical gap: compares orders of magnitude only.

	a	f_B	$f_{\inf}$			$f_{\sup}$			α	
	mm		meas.	PWE	z	meas.	PWE	z	meas.	PWE
ΓX	32	5367	3806(84)	3892	1.0	5920(190)	6037	0.6	0.81(39)	5.4
	40	4294	3590(150)	3387	1.3	5120(230)	4752	1.6	1.98(41)	4.0
	48	3578	2970(100)	2988	0.2	4230(140)	3880	2.6	1.45(29)	2.8
	56	3067	2917(62)	2666	3.9	3720(120)	3271	3.6	1.53(11)	1.9
ΓM	32	7584	5500(270)	6180	2.5	—†	6990	—	0.73(20)	0.1‡
	40	6067	4560(170)	5109	3.1	7150(410)	5818	3.2	0.81(28)	0.5‡
	48	5056	4040(93)	4385	3.6	—†	4942	—	1.30(17)	0.6‡

5 Discussion

Position of the gap along ΓX. The theory places the gap below f_B (from $0.93\,f_B$ at $a = 32$ mm to $0.97\,f_B$ at $a = 56$ mm), because the air flowing around the bricks behaves like a slower medium. Our data show a similar trend, but about four times stronger. In contrast, for $a = 56$ mm both edges are 9–14 % above the prediction, and we did not find the cause. Most of the failure of the line through the origin comes from this stronger trend, even though its slope is close to c.

Attenuation along ΓX. However, the slope is only 15 to 80 % of the predicted α, and D never goes much above 15 dB. We think some sound reaches the microphone without crossing the rows, either through small gaps between the LEGO bricks and the cardboard or around the enclosure. If this sound is about 15 dB weaker than the incident sound, D cannot become larger than that, and a straight line fitted to the data has a smaller slope. The larger the predicted α, the sooner D reaches this limit, which agrees with the measured/predicted ratio going from 15 % at $a = 32$ mm to 80 % at $a = 56$ mm.

Position of the gap along ΓM. Above f_B, the rows act as a diffraction grating and send part of the sound sideways, which probably explains why the gap does not close. A possible explanation for the low lower edge is that the crystal is only about two wavelengths wide. Because of this, the sound inside it spreads over angles of about $\lambda/W \approx 30°$, and the part that travels closer to ΓX may already be blocked by the ΓX gap: the measured $f_{\inf}$ along ΓM (1.02–1.13 times f_B along ΓX) lies inside the ΓX gap for the same a. However, our data cannot confirm this.

Attenuation along ΓM. Along ΓM the side walls were made of foam instead of LEGO, so sound could go around the crystal more easily, which makes the gap shallower. The comparison with the theory is rough here: the theoretical gap covers only 14–38 % of the $f_B \pm 10\,\%$ window where D is measured, and the theory predicts almost no attenuation there. Therefore, the measured attenuation must come mostly from other effects, such as sound crossing the crystal at an angle.

Finite crystal. The theory we used is for an infinite crystal. The gap needs a few rows to form: for $N \leq 2$ we almost never see it, so we only average $N \geq 4$. Along ΓX the LEGO walls reflect the sound like mirrors, so the 4 columns behave roughly like a wider crystal. Along ΓM the foam does not reflect the sound, so the crystal is far from infinite.

Limitations. The lattice constant a can only change in steps of 8 mm, and the columns are only 3.8 cm tall and not perfectly smooth or sealed. Background noise may also be a problem: the measurements were usually made with loud noise from other experiments in the lab (this effect is reduced because there were 5 recordings for each configuration). As mentioned in the analysis, the largest source of error is the reference.

6 Conclusion

A LEGO crystal has a Bragg gap along ΓX whose lower edge agrees with the PWE within 1.3σ for $a \leq 48$ mm, and which does not scale as $1/a$. The attenuation is linear in N, limited to ≈ 15 dB by sound that goes around the crystal. Along ΓM the gap appears about $\sqrt{2}$ higher, but the lower edge is 8–11 % below the PWE and the gap does not close; there is no evidence of a complete gap.

A Additional data and measurements

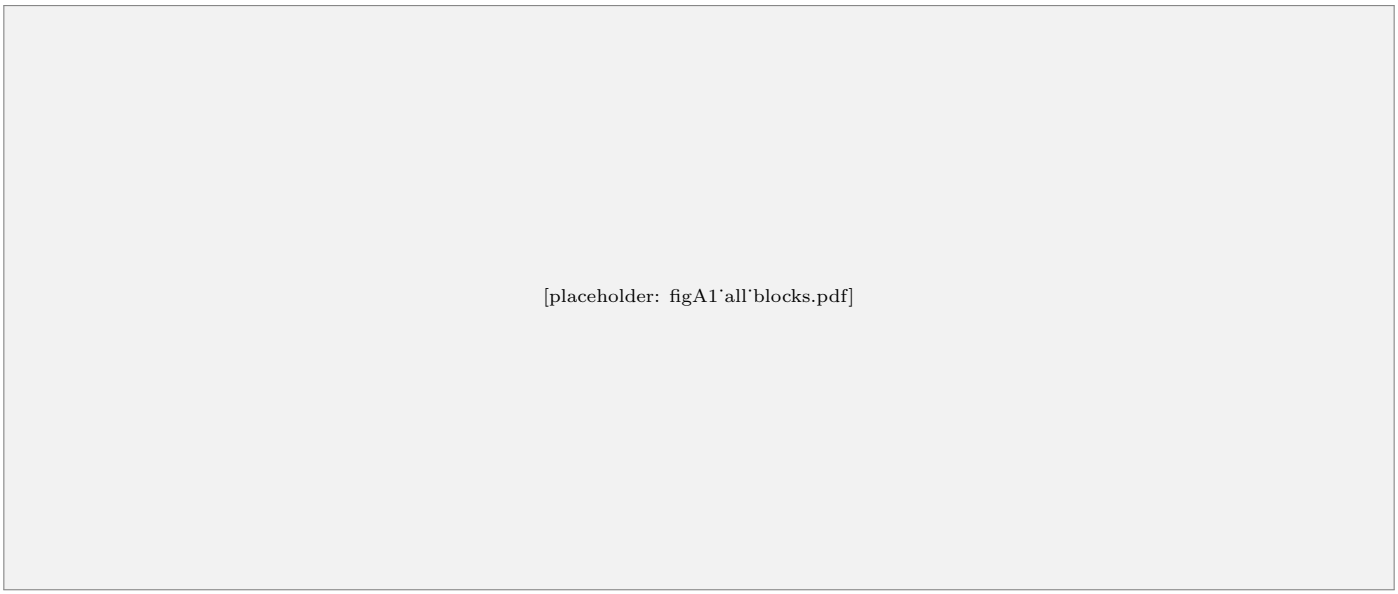


Figure 4. $T - T_0$ for all blocks with $N \geq 4$ (top: ΓX; bottom: ΓM). Band: PWE gap; red dashed: onset of lateral diffraction, c/L; dotted: threshold $T_0 - 6\,\mathrm{dB}$. Along ΓM there are no measurements with $a = 56\,\mathrm{mm}$ and $N \geq 4$.



Figure 5. f_{inf} and f_c (mean over $N \geq 4$ for each a) versus $x = 1/(2d)$, with the line through the origin and the free line; below, normalised residuals. Dashed: $f = c\,x$.



Figure 6. Top: ΓX ($N = 10$) and ΓM ($N = 11$) for the same a, with the measured gaps (bands) and f_B (dashed). Bottom: $f_{\rm inf}/f_B$ and f_c/f_B along both directions.

Table 2. All measurements. "rip.": ripple of T in the control band (dB); "$k/5$ rep.": repetitions used; "ref. other a": reference from the same session but from another block. Total uncertainty $\sigma^2 = \sigma^2_{\text{est}} + \sigma^2_{\text{rip}} + \sigma^2_{\text{ref}}$ ($+\sigma^2_{\text{res}}$). The components are also in `tabela.csv`.

dir	a (mm)	N	day	reference	rip.	D (dB)	f_{inf} (Hz)	f_{sup} (Hz)	notes
ΓX	32	1	10-01	ref_a32	4.2	-1.3 ± 4.3	—	—	
ΓX	32	2	10-01	ref_a32	6.2	0.1 ± 6.0	—	—	
ΓX	32	3	10-01	ref_a32	7.6	3.5 ± 7.2	5529 ± 889	5609 ± 599	
ΓX	32	4	10-01	ref_a32	1.7	11.4 ± 2.1	3947 ± 195	5768 ± 1747	4/5 rep.
ΓX	32	5	10-01	ref_a32	1.8	8.0 ± 2.7	3846 ± 133	5622 ± 161	
ΓX	32	6	10-01	ref_a32	2.2	12.7 ± 2.8	3795 ± 67	5802 ± 109	
ΓX	32	7	10-01	ref_a32	1.7	13.5 ± 2.3	3781 ± 70	6431 ± 165	
ΓX	32	8	10-01	ref_a32	2.1	13.4 ± 2.8	3880 ± 91	6028 ± 181	
ΓX	32	9	10-01	ref_a32	1.5	9.9 ± 1.8	3794 ± 79	5981 ± 893	
ΓX	32	10	10-01	ref_a32	1.6	13.1 ± 2.0	3773 ± 78	5890 ± 919	4/5 rep.
ΓX	40	1	10-06	ref_a56	3.1	-4.2 ± 3.1	—	—	4/5 rep., ref. other a
ΓX	40	2	10-06	ref_a56	5.4	3.9 ± 5.1	4538 ± 944	4713 ± 254	ref. other a
ΓX	40	3	10-06	ref_a56	1.0	6.0 ± 1.2	3196 ± 330	4832 ± 126	4/5 rep., ref. other a
ΓX	40	4	10-06	ref_a56	1.2	8.9 ± 1.6	3474 ± 168	5677 ± 114	ref. other a
ΓX	40	5	10-06	ref_a56	4.2	7.2 ± 4.1	3625 ± 327	4783 ± 163	ref. other a
ΓX	40	6	10-06	ref_a56	2.0	14.5 ± 2.0	3525 ± 122	4818 ± 271	ref. other a
ΓX	40	7	10-01	ref_a40	4.4	19.0 ± 5.3	3696 ± 157	4875 ± 185	
ΓX	40	8	10-01	ref_a40	4.7	10.1 ± 4.8	3833 ± 213	5510 ± 462	
ΓX	40	9	10-01	ref_a40	1.9	13.2 ± 3.2	3669 ± 330	4854 ± 111	
ΓX	40	10	10-01	ref_a40	2.2	18.2 ± 4.1	3556 ± 101	—	does not close
ΓX	48	1	10-06	ref_a48	1.9	-1.1 ± 2.7	—	—	
ΓX	48	2	10-06	ref_a48	2.4	2.5 ± 2.7	—	—	4/5 rep.
ΓX	48	3	10-06	ref_a48	2.3	4.6 ± 2.7	3600 ± 502	3770 ± 1598	
ΓX	48	4	10-06	ref_a48	2.3	5.2 ± 2.9	3613 ± 568	4173 ± 364	
ΓX	48	5	10-06	ref_a48	2.7	8.7 ± 2.8	2883 ± 125	4157 ± 260	
ΓX	48	6	10-06	ref_a48	2.5	9.6 ± 2.9	2970 ± 100	4071 ± 161	4/5 rep.
ΓX	48	7	10-06	ref_a48	2.0	9.4 ± 2.6	2982 ± 98	4093 ± 130	4/5 rep.
ΓX	48	8	10-06	ref_a48	2.0	10.7 ± 2.1	2974 ± 83	4306 ± 112	
ΓX	48	9	10-06	ref_a48	2.3	9.6 ± 2.8	3019 ± 79	4332 ± 96	
ΓX	48	10	10-06	ref_a48	2.8	15.0 ± 2.7	2850 ± 135	4210 ± 123	
ΓX	56	1	10-06	ref_a56	0.8	0.7 ± 0.7	—	—	
ΓX	56	2	10-06	ref_a56	1.1	2.7 ± 1.0	—	—	4/5 rep.
ΓX	56	3	10-06	ref_a56	0.5	4.4 ± 0.5	3018 ± 50	3142 ± 230	
ΓX	56	4	10-06	ref_a56	1.1	5.7 ± 1.0	2993 ± 75	3418 ± 200	
ΓX	56	5	10-06	ref_a56	1.6	9.1 ± 1.5	2912 ± 72	3607 ± 133	4/5 rep.
ΓX	56	6	10-06	ref_a56	1.6	10.6 ± 1.5	2941 ± 56	3587 ± 98	4/5 rep.
ΓX	56	7	10-06	ref_a56	0.8	11.3 ± 0.7	2884 ± 46	3593 ± 77	
ΓX	56	8	10-06	ref_a56	1.3	11.0 ± 1.2	2979 ± 47	3993 ± 103	
ΓX	56	9	10-06	ref_a56	1.5	15.1 ± 1.5	2595 ± 266	3809 ± 215	
ΓX	56	10	10-06	ref_a56	1.0	13.3 ± 0.9	2821 ± 58	3886 ± 85	
ΓM	32	2	10-13	ref_a32	1.6	3.3 ± 1.5	6288 ± 297	6883 ± 150	
ΓM	32	3	10-13	ref_a32	1.7	4.0 ± 1.6	5950 ± 96	6953 ± 128	
ΓM	32	4	10-13	ref_a32	2.0	3.9 ± 1.8	7871 ± 1018	8223 ± 817	4/5 rep.
ΓM	32	5	10-13	ref_a32	2.0	2.7 ± 1.9	6005 ± 2114	6989 ± 120	
ΓM	32	6	10-13	ref_a32	2.0	7.4 ± 1.9	6076 ± 179	—	does not close
ΓM	32	7	10-13	ref_a32	1.8	8.4 ± 1.6	5910 ± 73	—	4/5 rep., does not close
ΓM	32	8	10-13	ref_a32	1.6	10.0 ± 1.5	4603 ± 189	—	does not close
ΓM	32	9	10-13	ref_a32	1.9	5.7 ± 1.8	4680 ± 206	8281 ± 309	
ΓM	32	10	10-13	ref_a32	1.9	8.7 ± 1.8	4581 ± 141	—	does not close
ΓM	32	11	10-13	ref_a32	2.4	9.2 ± 2.2	4664 ± 353	—	does not close
ΓM	40	1	10-13	ref_a40_fim	5.3	-2.1 ± 4.9	—	—	
ΓM	40	2	10-13	ref_a40_fim	1.4	4.8 ± 1.3	6260 ± 302	6794 ± 1291	
ΓM	40	3	10-13	ref_a40_fim	1.9	6.1 ± 1.7	4712 ± 335	5756 ± 593	4/5 rep.
ΓM	40	4	10-13	ref_a40_fim	1.4	7.7 ± 1.3	4633 ± 101	—	4/5 rep., does not close
ΓM	40	5	10-13	ref_a40_fim	1.9	10.2 ± 1.7	4695 ± 445	7151 ± 209	
ΓM	40	6	10-13	ref_a40_fim	1.9	12.4 ± 1.7	4706 ± 213	8699 ± 493	
ΓM	40	7	10-13	ref_a40_fim	2.3	11.4 ± 2.2	4535 ± 210	6865 ± 1507	
ΓM	40	8	10-13	ref_a40_fim	2.0	14.1 ± 1.8	4544 ± 175	8655 ± 429	
ΓM	40	9	10-13	ref_a40_fim	2.5	11.1 ± 2.3	4624 ± 342	—	does not close
ΓM	40	10	10-13	ref_a40_fim	2.6	7.4 ± 2.4	4616 ± 184	—	does not close
ΓM	40	11	10-13	ref_a40_fim	2.6	9.9 ± 2.4	4206 ± 165	6696 ± 177	4/5 rep.
ΓM	48	1	10-12	ref_a48_fim	1.4	0.4 ± 1.5	—	—	
ΓM	48	2	10-12	ref_a48_fim	1.3	4.2 ± 1.8	5838 ± 656	—	4/5 rep.
ΓM	48	3	10-12	ref_a48_fim	1.2	6.5 ± 1.7	3896 ± 112	—	
ΓM	48	4	10-12	ref_a48_fim	1.3	8.3 ± 1.6	4148 ± 215	6287 ± 557	
ΓM	48	5	10-12	ref_a48_fim	1.9	8.5 ± 2.3	4131 ± 176	6232 ± 769	
ΓM	48	6	10-12	ref_a48_fim	1.0	9.6 ± 1.2	3966 ± 61	—	does not close
ΓM	48	7	10-12	ref_a48_fim	1.7	7.0 ± 2.2	4314 ± 76	5068 ± 144	
ΓM	48	8	10-12	ref_a48_fim	2.7	8.7 ± 2.7	4360 ± 236	6941 ± 973	
ΓM	48	9	10-12	ref_a48_fim	0.9	13.6 ± 1.3	3945 ± 50	—	does not close
ΓM	48	10	10-12	ref_a48_fim	1.2	14.8 ± 1.4	3990 ± 85	—	does not close
ΓM	48	11	10-12	ref_a48_fim	2.1	12.2 ± 3.0	4260 ± 188	—	does not close

B Calibrations and derivations

Speed of sound.. $c(\theta) = 331.3\sqrt{1+\theta/273.15}\,\mathrm{m/s}$, $\sigma_c = c\,\sigma_\theta/[2(273.15+\theta)] = 0.6\,\mathrm{m/s}$ for $\sigma_\theta = 1\,^\circ\mathrm{C}$.

Plane-wave Criterion.. The loudspeaker emits approximately spherical waves. For the wave at the crystal entrance to be treated as plane, the distance r from the loudspeaker to the entrance must be large compared with the entrance width W. The loudspeaker faces the centre of the entrance, so its edges are at $W/2$ from the axis and are reached after an extra path, relative to the centre, of $\delta = \sqrt{r^2+(W/2)^2} - r \approx W^2/(8r)$. This extra path corresponds to a phase difference of $360^\circ\,\delta/\lambda$ across the entrance, and we require $\delta \le \lambda/10$ (at most 36°), which gives

$$r \ge \frac{1.25\,W^2}{\lambda_{\min}}, \tag{4}$$

where $\lambda_{\min} = c/f_{\max}$ is the smallest wavelength used in the analysis: δ does not depend on frequency, but λ decreases with it, so the condition is most demanding at the highest frequency. The highest frequency we use is the upper gap edge, which in Table 1 is at most $1.21\,f_B$ ($a = 56\,\mathrm{mm}$), so we take $f_{\max} = 1.2 f_B$, i.e. $\lambda_{\min} = 2d/1.2$. Along ΓX, $W = 18$ studs $= 144\,\mathrm{mm}$ for all a; along ΓM, $W = 14$, 17 and 20 studs (112, 136 and 160 mm) for $a = 32$, 40 and 48 mm.

Table 3. Minimum distance $r_{\min} = 1.25\,W^2/\lambda_{\min}$, with $\lambda_{\min} = 2d/1.2$ (upper edge of the gap), and margin with respect to $r = 70\,\mathrm{cm}$.

Lattice	a (mm)	d (mm)	W (mm)	$\lambda_{\min}$ (mm)	$r_{\min}$ (cm)	70 cm/$r_{\min}$
ΓX	32	32.0	144	53.3	48.6	1.44
ΓX	40	40.0	144	66.7	38.9	1.80
ΓX	48	48.0	144	80.0	32.4	2.16
ΓX	56	56.0	144	93.3	27.8	2.52
ΓM	32	22.6	112	37.7	41.6	1.68
ΓM	40	28.3	136	47.1	49.0	1.43
ΓM	48	33.9	160	56.6	56.6	1.24

In all configurations $r = 70\,\mathrm{cm}$ exceeds $r_{\min}$, with a minimum margin of 1.24 (ΓM, $a = 48\,\mathrm{mm}$).

Plane-wave expansion (PWE).. With $\sigma = 1/\rho$, $\eta = 1/(\rho c^2)$ and $p(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}\sum_{\mathbf{G}} p_{\mathbf{G}}\, e^{i\mathbf{G}\cdot\mathbf{r}}$, $\mathbf{G} = \frac{2\pi}{a}(m,n)$, the wave equation gives, for each $\mathbf{k}$, the eigenvalue problem

$$\sum_{\mathbf{G}'} \sigma(\mathbf{G}-\mathbf{G}')\,(\mathbf{k}+\mathbf{G})\cdot(\mathbf{k}+\mathbf{G}')\,p_{\mathbf{G}'} = \omega^2 \sum_{\mathbf{G}'} \eta(\mathbf{G}-\mathbf{G}')\,p_{\mathbf{G}'},$$

with, for a square rod of side s, $\sigma(\mathbf{G}) = (\sigma_{\mathrm{t}} - \sigma_{\mathrm{air}})\,ff\,\mathrm{sinc}(G_x s/2)\,\mathrm{sinc}(G_y s/2)$ ($\mathbf{G} \neq 0$). We use $15^2 = 225$ waves (729 waves change the edges by 0.1–0.4 %). Only modes symmetric with respect to the plane containing the sound direction count (the antisymmetric ones are not excited by a wave arriving head-on). The gap is $f_{\inf} = \max_k f_1(k)$, $f_{\sup} = \min_k f_{n\ge 2}(k)$, with k from Γ to X ($|\mathbf{k}| = \pi/a$) or from Γ to M ($|\mathbf{k}| = \sqrt{2}\pi/a$). The attenuation comes from the inverse problem (given ω, a complex k comes out), $(k^2 M_2 + k M_1 + M_0)\,\mathbf{p} = 0$, linearised into a problem of double dimension; κ is the smallest $|\,\mathrm{Im}\,k|$. With only the two waves $e^{\pm i\pi x/d}$, $f_{\sup} - f_{\inf} \propto |\sigma(2\pi/d)|$: the gap widens with the amount of brick per plane.

Effective medium.. At low frequency, $\kappa_{\mathrm{eff}} = (1-ff)\kappa$ and $\rho_{\mathrm{eff}} \simeq \rho(1+ff)/(1-ff)$ (rigid rods), hence $c_{\mathrm{eff}} = c/\sqrt{1+ff}$ and $f_{\mathrm{gap}} \approx f_B/\sqrt{1+ff} = 0.90,\ 0.93,\ 0.95,\ 0.96\,f_B$ for $a = 32$–$56\,\mathrm{mm}$.

Complete gap.. There is a gap along all directions if $f_{\sup}(\Gamma X) > f_{\inf}(\Gamma M)$. Along ΓM the diagonal of the brick (22.3 mm) occupies almost the whole distance between planes ($a/\sqrt{2} = 22.6\,\mathrm{mm}$ for $a = 32\,\mathrm{mm}$): the medium looks almost uniform and the ΓM gap is narrow, so the predicted ratio $f_{\inf}(\Gamma M)/f_{\inf}(\Gamma X)$ is 1.59, 1.51, 1.47 (and not $\sqrt{2}$) and there is no overlap for 2×2 bricks ($-7\,\mathrm{Hz}$ even for $a = 24\,\mathrm{mm}$).

Uncertainty of the depth.. $D = T_0 - M$; with $n_{\mathrm{base}} = \log_2(0.70/0.35)/(1/6) = 6$ and $n_{\mathrm{win}} = \log_2(1.1/0.9)/(1/6) = 1.74$ independent points, and the variance of the median being $\pi/2$ times that of the mean, $\mathrm{Var}\,D = \left(\frac{\pi}{2}\frac{1}{6} + \frac{1}{1.74}\right)\sigma^2$, i.e. $\sigma_{\mathrm{rip}}(D) = 0.92 \times$ ripple. The dispersion of the references uses weights $w_k = 1/\mathrm{rip}_k^2$: $\sigma_{\mathrm{ref}}^2 = \sum_k w_k (x_k - x_0)^2 / \sum_k w_k$.

Fits.. For $y = mx + b$ with $w_i = 1/\sigma_i^2$: $m = (S\,S_{xy} - S_x S_y)/\Delta$, $b = (S_{xx} S_y - S_x S_{xy})/\Delta$, $\sigma_m^2 = S/\Delta$, $\sigma_b^2 = S_{xx}/\Delta$, $\Delta = S\,S_{xx} - S_x^2$; the uncertainty in x enters through $\sigma_{i,\mathrm{eff}}^2 = \sigma_{y,i}^2 + (m\sigma_{x,i})^2$, minimising $\chi^2(m,b)$ numerically. If $\chi^2/\nu > 1$, $\sigma \to \sqrt{\chi^2/\nu}\,\sigma$; uncertainties are never shrunk. The regressions were checked against `numpy.polyfit`, `scipy.optimize.curve_fit` and `scipy.odr`.

C Code

Repository: https://github.com/FM767/LFE-acoustics.

- `cristal_simples/` and `cristal_lego_g/cristal_lego_g/`: measurement apps (ΓX and ΓM; Python, `sounddevice`) with the measurement plan, quality control and the recordings (`projeto/dados/<step>/*.wav` + `meta.json`; repeated measurements in `_antigos/`, which do not count).
- `cristal_lego_g/analise 2.0/`: joint analysis of the two directions. `analise.py` (spectra, $T(f)$, quantities, uncertainties, fits), `teoria.py` (PWE), `graficos.py`, `app.py` (interface); results in `resultados/` (`resultados.txt`, `tabela.csv`, `ajustes.csv`, `teoria.txt`). All equations, in order, in `EQUACOES.md`.
- `relatorio/`: this report. `fazer_figuras.py` generates Figs. 2–3 and Table A1 from `dados_figuras/`; `dados_figuras/curvas_T.py` recomputes the curves of Fig. 2 from the `.wav` files (same method, numpy only).

Parameters: FS = 48000, NPERSEG = 4096, band 0.5–12 kHz, segment 0.8–10.3 µs, 1/6-octave smoothing, 6 dB threshold, $\sigma_a = 0.1\,\mathrm{mm}$, $\sigma_\theta = 1\,^\circ\mathrm{C}$.

D Use of AI

We used Claude (Anthropic) as an assistant: in planning the experiment and choosing the parameters, in writing the measurement and analysis code (which we reviewed and tested: the regressions were compared with independent libraries and the coverage of the uncertainties by simulation) and in drafting this report from the results files. The set-up, the measurements, the decisions on what to measure and the verification of the numbers against `resultados.txt` and `teoria.txt` were done by us.